

CST8912 – Cloud Solution Architecture

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Graded Lab Activity #3

Introduction:

As part of this lab, we examine how to manage and configure Azure Storage accounts. We focus on important tasks like configuring blob access tiers, enabling secure file sharing using shared access signatures (SAS), and selecting redundancy choices. Likewise, we also use lifecycle management to automate data transfer among various storage levels according to preset rules.

The following activities are done hands-on in the lab exercises:

- Initiating a geo-redundant storage account (GRS).
- Adjusting blob access levels and storage redundancy.
- File uploading, access control management, and container creation.
- Providing temporary, secure access to storage resources by generating SAS tokens.
- putting lifecycle management guidelines into practice to automate data transfer.

Prerequisites:

Any sample file can be selected for this lab from the link given below:

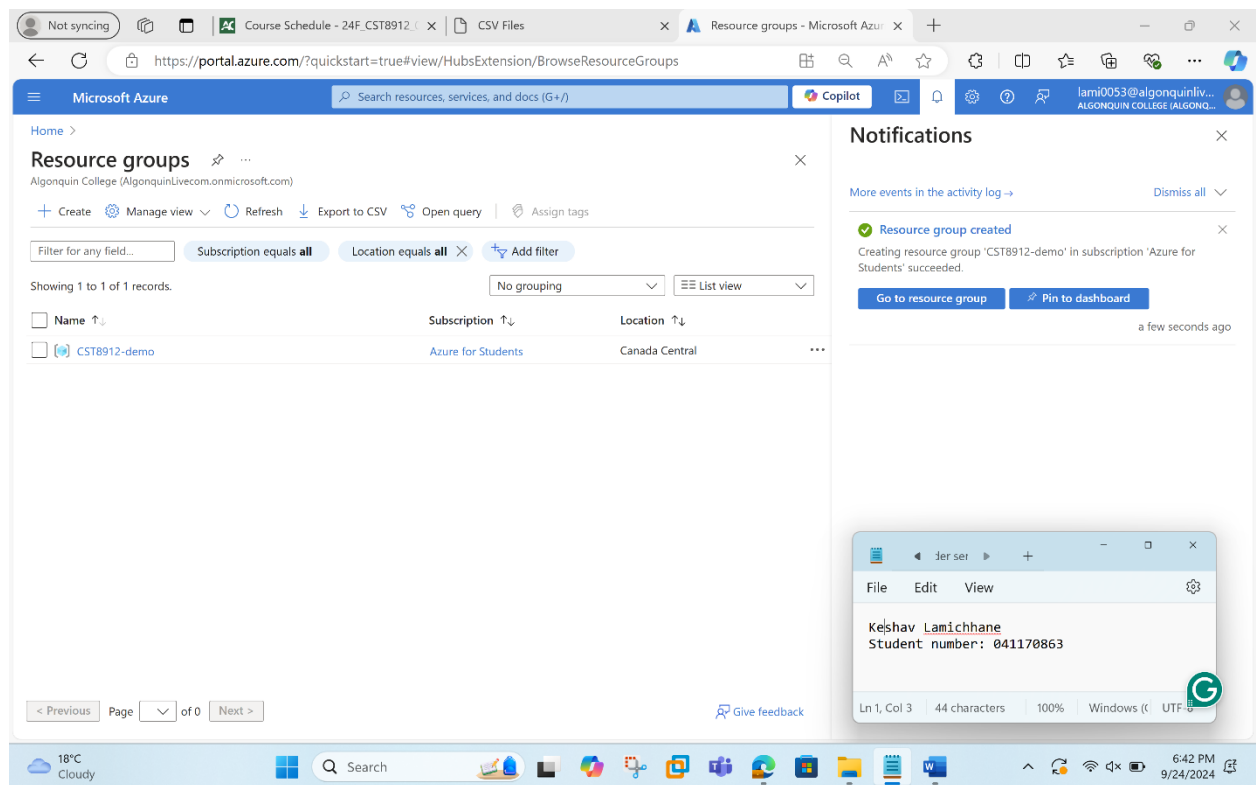
<https://people.sc.fsu.edu/~jburkardt/data/csv/csv.html>

Objectives:

- Define and list the essential architectural elements required to satisfy the functional needs of various application scenarios.
- Implement highly available solutions by gaining a profound understanding of cloud service models, such as IaaS, PaaS, and SaaS.
- Understand how to use Shared Access Signatures (SAS) and learn how to design and oversee a highly scalable, secure cloud storage solution.

Findings: The findings of this lab can be summarized in following steps:

Step 1: This step involves selecting geo-redundant storage keeping networking and data protection settings as they are and creating a storage account under the student subscription and resource group "CST8912-demo" for area Canada Central.



In the next step, we created the storage named lab2test8912 with the following details:

Subscription: Select your student subscription.

Resource group: Select or create a new one named "CST8912-demo."

Storage account name: lab2test8912

Region: Select "Canada Central."

Performance: Standard.

Redundancy: Geo-redundant storage (GRS).

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Not syncing | Course Schedule - 24F_CST8912... | CSV Files | Create a storage account - Micro... | +

https://portal.azure.com/?quickstart=true#create/Microsoft.StorageAccount-ARM

Microsoft Azure | Search resources, services, and docs (G+)

Home > Storage accounts >

Create a storage account

Project details
Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription * | Azure for Students |
Resource group * | CST8912 demo |
[Create new](#)

Instance details

Storage account name * | lab2test8912 |
Region * | (Canada) Canada Central |
[Deploy to an Azure Extended Zone](#)
Primary service | Select a primary service |
Performance * | ☒ Standard: Recommended for most scenarios (general-purpose v2 account) |
☐ Premium: Recommended for scenarios that require low latency.
Redundancy * | Geo-redundant storage (GRS) |
☒ Make read access to data available in the event of regional unavailability.

[Previous](#) | [Next](#) | [Review + create](#)

Keshav Lamichhane
Student number: 041170863

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Step 2: In this step, we navigate to the "Storage accounts" service and click on lab2test8912. In the left-hand menu under "Data management," we change the redundancy to local redundant storage and also change the blob access tier to cool.

Microsoft Azure

Home > Storage accounts > lab2test8912

Storage accounts

Algonquin College (AlgonquinLivecom.onmicrosoft.com)

+ Create ? Restore ...

Filter for any field...

Name ↑

lab2test8912

lab2test8912 | Redundancy

Storage account

Search Save Discard Prepare for failover Refresh Give feedback

▼ Data storage

- Containers
- File shares
- Queues
- Tables

▼ Security + networking

- Networking
- Front Door and CDN
- Access keys
- Shared access signature
- Encryption
- Microsoft Defender for Cloud

▼ Data management

- Storage tasks (preview)
- Redundancy**
- Data protection
- Object replication
- Blob inventory

Redundancy

Azure Storage redundancy copies your data so that it is protected from transient hardware failures, network or power outages, and natural disasters. If an outage renders the primary endpoint unavailable, then you can initiate a failover to the secondary endpoint to rapidly restore write access to your data. [Learn more about storage account failover](#)

Last failover time -

Storage endpoints [View all](#)

Location	Data center type	Status
Canada Central	Primary	Available
Canada East	Secondary	Available

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Under the "Settings" section, click "Configuration." And in the "Blob access tier" section, change the default to "Cool".

Microsoft Azure

Home > Storage accounts > lab2test8912

Storage accounts

Algonquin College (AlgonquinLivecom.onmicrosoft.com)

+ Create ? Restore ...

Filter for any field...

Name ↑

lab2test8912

lab2test8912 | Configuration

Storage account

Search Save Discard Refresh Give feedback

Access keys

- Shared access signature
- Encryption
- Microsoft Defender for Cloud

▼ Data management

- Storage tasks (preview)
- Redundancy
- Data protection
- Object replication
- Blob inventory

▼ Settings

- Configuration**
- Data Lake Gen2 upgrade
- Resource sharing (CORS)
- Advisor recommendations
- Endpoints
- Locks
- Monitoring
- Monitoring (classic)
- Automation
- Help

Secure transfer required ☒ Disabled ☒ Enabled

Allow Blob anonymous access ☒ Disabled ☐ Enabled

Allow storage account key access ☐ Disabled ☒ Enabled

Allow recommended upper limit for shared access signature (SAS) expiry interval ☒ Disabled ☐ Enabled

Default to Microsoft Entra authorization in the Azure portal ☒ Disabled ☐ Enabled

Minimum TLS version

Permitted scope for copy operations (preview)

Blob access tier (default) ☐ Hot ☒ Cool

Large file shares ☐ Disabled ☒ Enabled

Successfully updated storage account 'lab2test8912'. Settings may take up to 30 seconds to take effect.

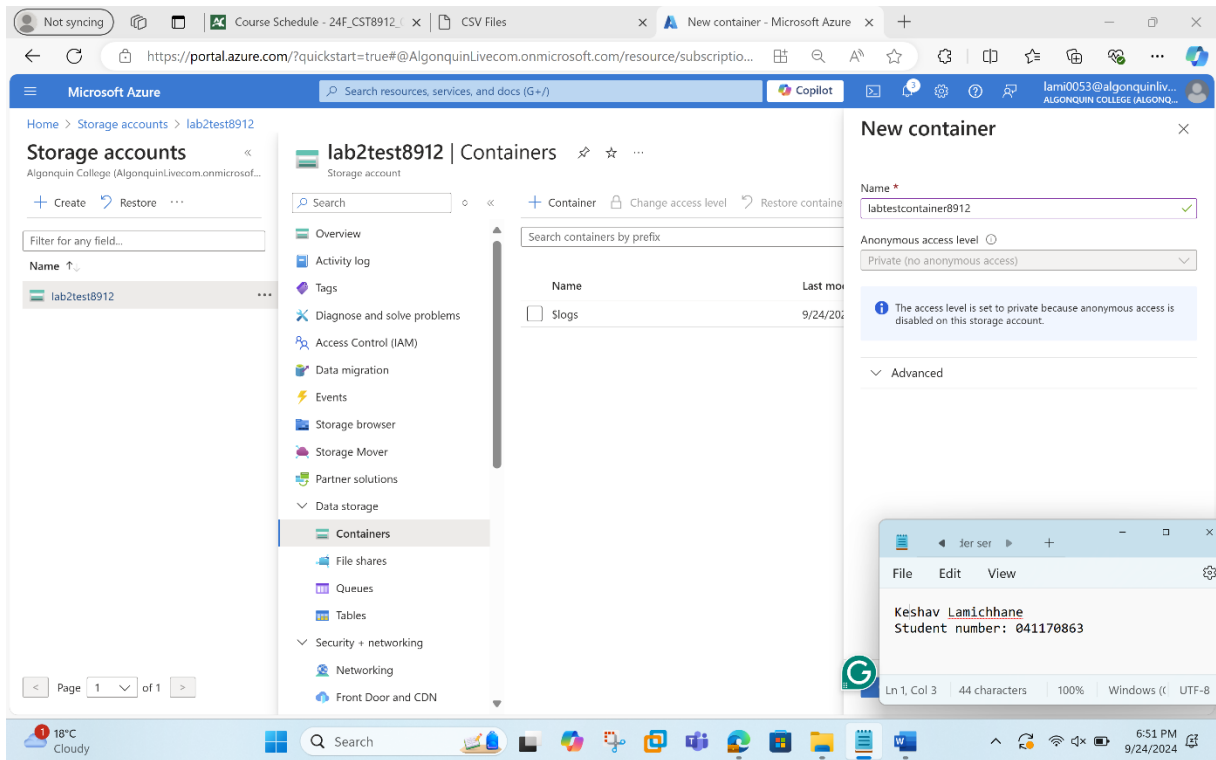
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Step 3: In this step, we will be creating container named as labtestcontainer8912 and set the access to public and click create.



Next step is to upload a blob. For this, click on upload section of container created so far. The sample file for uploading can be downloaded from the link provided in the lab instructions.

Once uploading is done,

- Change "Access tier" to "Hot."
- Select the destination folder "sampletest8912."
- Click "Upload."

The following screenshot clarifies the step:

Microsoft Azure

Home > Storage accounts > lab2test8912 | Containers >

labtestcontainer8912

Container

Search

Upload Change access level Refresh Delete Change tier

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: labtestcontainer8912

Search blobs by prefix (case-sensitive)

Add filter

Name	Modified	Access tier	Access
No results			

Upload blob

1 file(s) selected: airtravel.csv

Drag and drop files here or [Browse for files](#)

☐ Overwrite if files already exist

Advanced

Upload Give feedback

File Edit View

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Microsoft Azure

Home > Storage accounts > lab2test8912 | Containers >

labtestcontainer8912

Container

Search

Upload Change access level Refresh Delete Change tier

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: labtestcontainer8912

Search blobs by prefix (case-sensitive)

Add filter

Name	Modified	Access tier	Access
No results			

Upload blob

Drag and drop files here or [Browse for files](#)

☐ Overwrite if files already exist

Advanced

Blob type

☒ Upload .vhd files as page blobs (recommended)

Block size

Access tier

Upload to folder

Blob index tags

Key	Value

Encryption scope

☒ Use existing default container scope

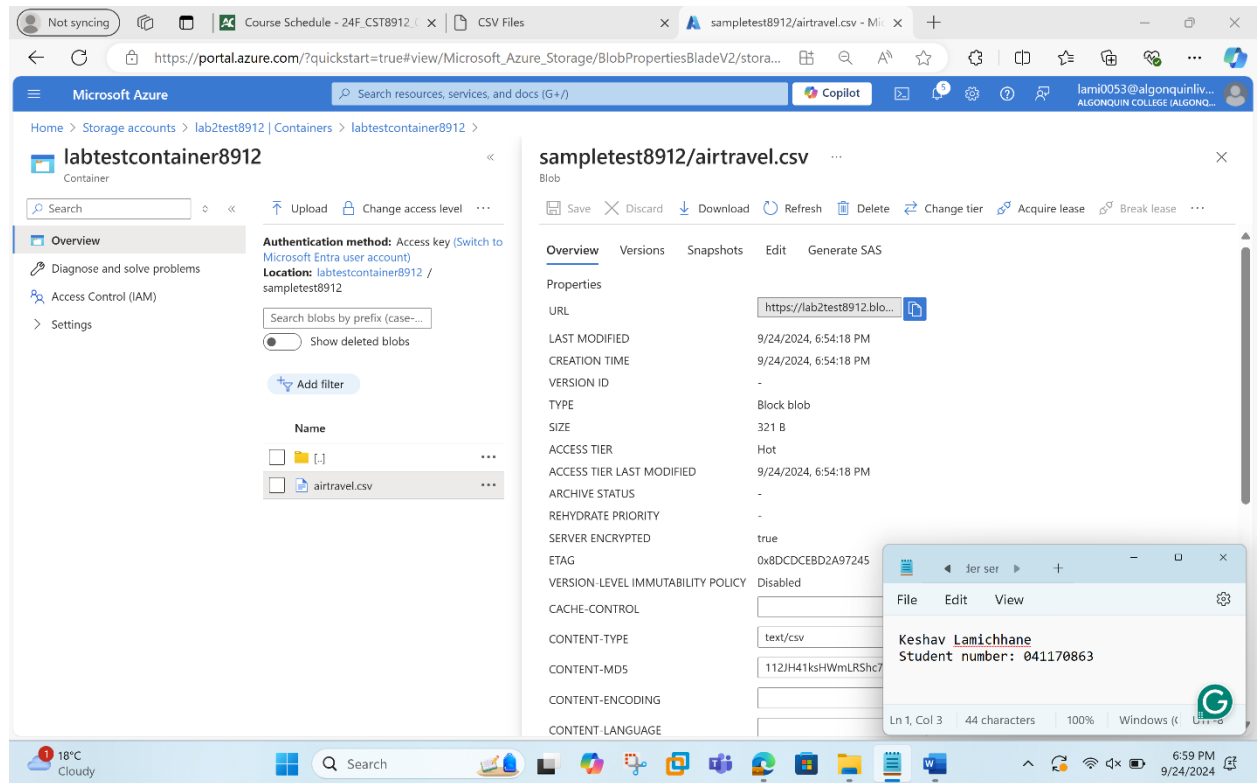
File Edit View

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Student number: 041170863

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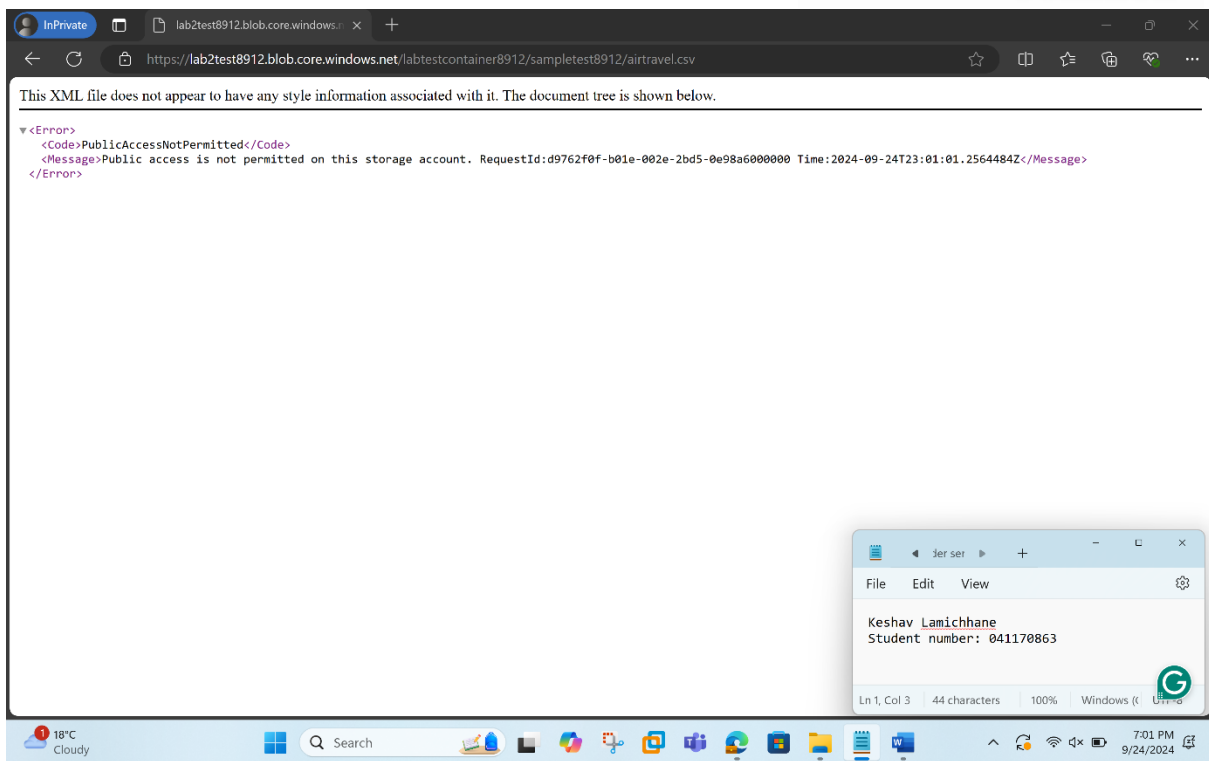
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Step 4: In this step, we see the configuration options of the container and copy the URL to open that in a new private window of the browser.

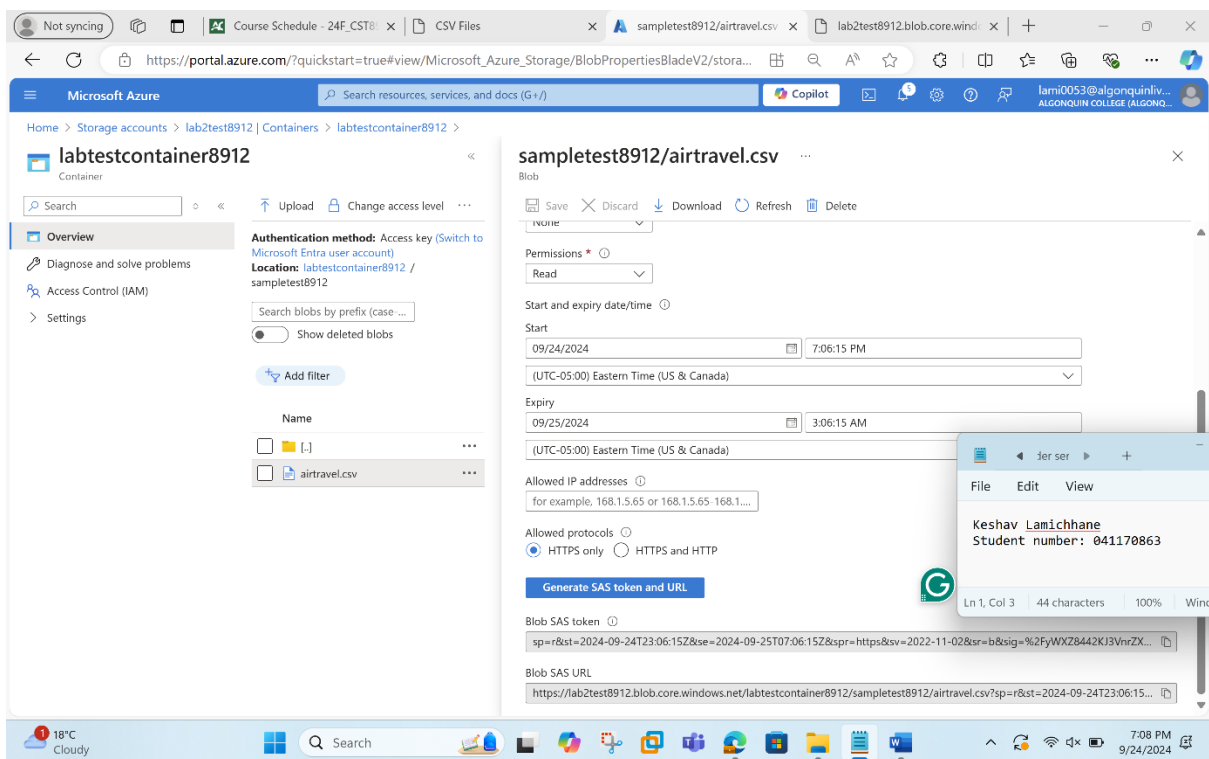


The screenshot shows the Microsoft Azure portal interface. On the left, the navigation pane shows the 'labtestcontainer8912' container. The main area displays the 'sampletest8912/airtravel.csv' blob. The 'Overview' tab is active, showing the blob's properties. The URL is highlighted as `https://labtest8912.blob.core.windows.net/airtravel.csv`. A small window in the bottom right corner displays the user's name 'Keshav Lamichhane' and student number '041170863'.

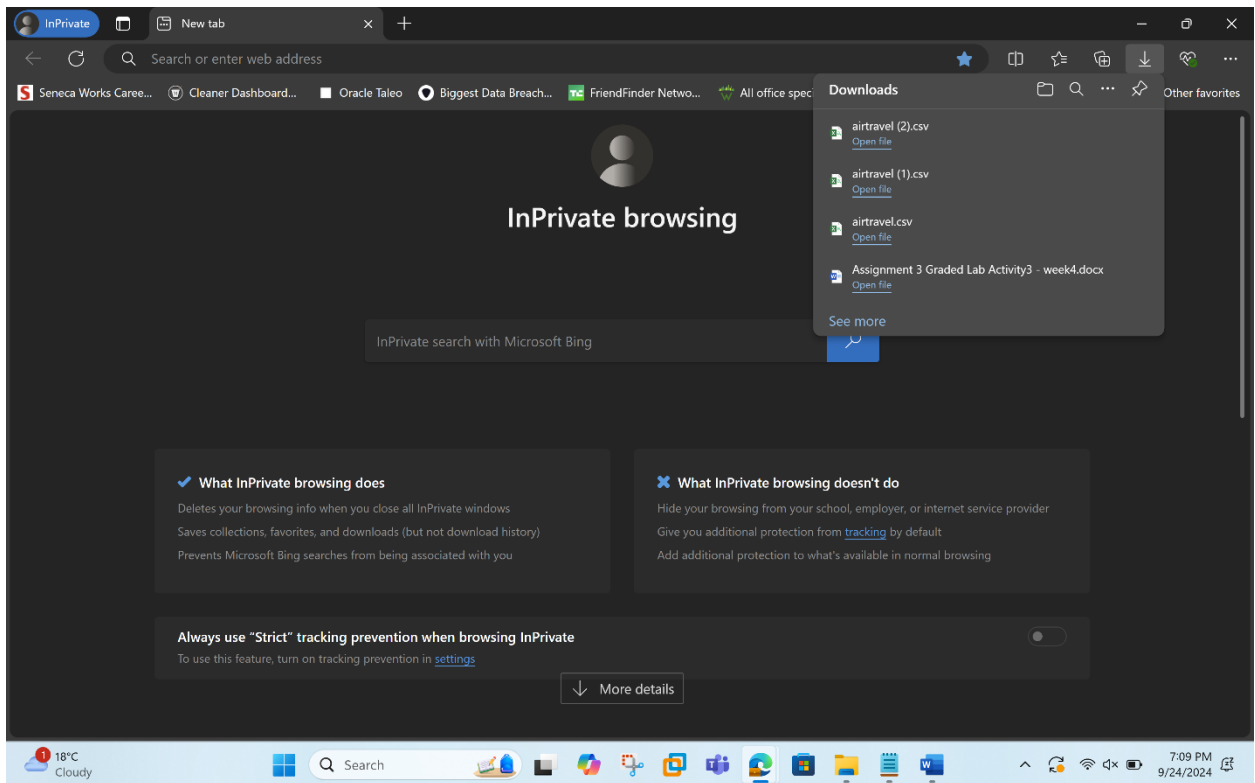
The url won't work since the container has been assigned private access and cannot be accessed by the public. We will receive a "Resource not found" error, indicating the blob is not publicly accessible.



Step 5: Now, we need to create SAS token and then try again pasting the blob SAS URL on the private window.



The file should now be accessible. As soon as you paste the link, the download will occur, and you can access the file.



Step 6: In this step, we perform the lifecycle management with the following setting:

Configure Lifecycle Management:

- Go back to the storage account's main page.
- Under "Data management," select "Lifecycle Management."
- Click "+ Add rule."
- Give the rule name as **myrule8912**.
- Set the "Rule scope" to "Limit blobs with filters."

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Not syncing | Course Schedule - 24F_CST@ | CSV Files | lab2test8912 - Microsoft Azure | lab2test8912.blob.core.windows.net

https://portal.azure.com/?quickstart=true#@AlgonquinLivecom.onmicrosoft.com/resource/subscription...

Microsoft Azure | Search resources, services, and docs (G+)

Home > Storage accounts > lab2test8912

Storage accounts

Algonquin College (AlgonquinLivecom.onmicrosoft.com)

+ Create + Restore ...

Filter for any field...

Name ↑

Name
lab2test8912

lab2test8912 | Lifecycle management

Search

+ Add a rule ✓ Enable ☐ Disable Refresh Delete Give feedback

Networking
Front Door and CDN
Access keys
Shared access signature
Encryption
Microsoft Defender for Cloud

Data management

- Storage tasks (preview)
- Redundancy
- Data protection
- Object replication
- Blob inventory
- Static website
- Lifecycle management**
- Azure AI Search

Settings

- Configuration

Lifecycle management offers a rich, rule-based policy for general purpose v2 and blob storage accounts. Use the policy to transition your data to the appropriate access tiers or expire at the end of the data's lifecycle. A new or updated policy may take up to 48 hours to complete. [Learn more](#)

List View Code View

Enable access tracking ☐

Name	Status	Blob type
No rules		

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- For the condition, set "Base blobs last modified" to 15 days and click on "Action," and select move to cool storage and finally save it.

Not syncing | Course Schedule - 24F_CST@ | CSV Files | Add a rule - Microsoft Azure | lab2test8912.blob.core.windows.net

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_Storage/AddRuleView.ReactView/stor...

Microsoft Azure | Search resources, services, and docs (G+)

Home > Storage accounts > lab2test8912 | Lifecycle management

Add a rule

Details Base blobs Filter set

A rule is made up of one or more conditions and actions that apply to the entire storage account. Optionally, specify that rules will apply to particular blobs by limiting with filters.

Rule name *

myrule8912

Rule scope *

☐ Apply rule to all blobs in your storage account

☒ Limit blobs with filters

Blob type *

☒ Block blobs

☐ Append blobs

Blob subtype *

☒ Base blobs

☐ Snapshots

☐ Versions

Previous Next

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Home > Storage accounts > lab2test8912 | Lifecycle management >

Add a rule

Details Base blobs Filter set

Lifecycle management uses your rules to automatically move blobs to cooler tiers or to delete them. If you create multiple rules, the associated actions must be implemented in tier order (from hot to cool storage, then archive, then deletion).

If

Base blobs were *

☒ Last modified

☐ Created

More than (days ago) *

15

Then

Move to cool storage

+ Add conditions

Previous Next

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Search

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Lifecycle management rule created is shown below:

Home > Storage accounts > lab2test8912

Storage accounts

Algonquin College (AlgonquinLivecom.onmicrosoft.com)

+ Create Restore ...

Filter for any field...

Name

lab2test8912

lab2test8912 | Lifecycle management

Storage account

Search

+ Add a rule Enable Disable Refresh Delete Give feedback

Lifecycle management offers a rich, rule-based policy for general purpose v2 and blob storage accounts. Use the policy to transition your data to the appropriate access tiers or expire at the end of the data's lifecycle. A new or updated policy may take up to 48 hours to complete. [Learn more](#)

List View Code View

Enable access tracking

Name	Status	Blob type
myrule8912	Enabled	Block

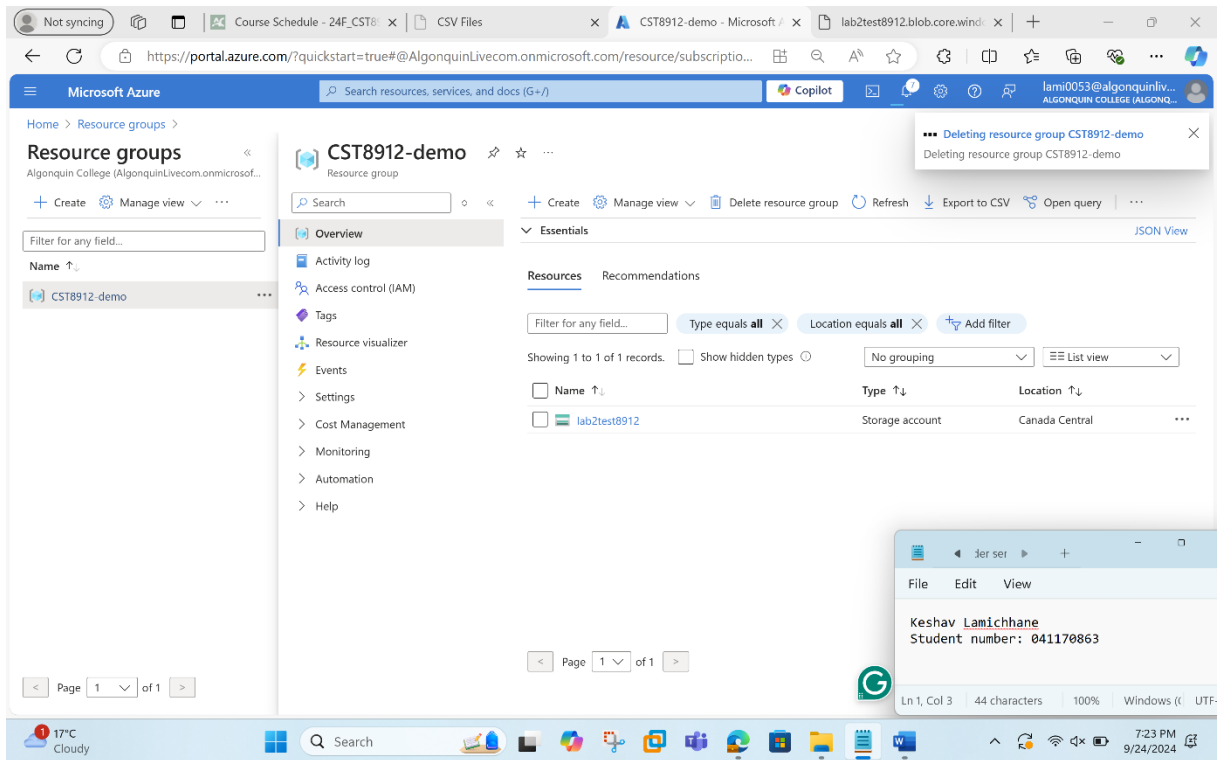
Page 1 of 1

17°C Cloudy

Search

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Step 7: This step concludes the lab. For proper cost management and resource management, we will be deleting the resources created so far for this lab.



The screenshot shows the Microsoft Azure portal interface. The main content area displays the 'CST8912-demo' resource group. Under the 'Resources' tab, a table lists the resources within the group:

Name	Type	Location
lab2test8912	Storage account	Canada Central

A notification at the top right indicates that the resource group 'CST8912-demo' is being deleted. In the bottom right corner, a small window displays the user's name 'Keshav Lamichhane' and student number '041170863'.

Takeaways and Analysis:

- Gained knowledge of setting up and creating Geo-Redundant Storage (GRS).
- Modified the blob access levels and storage account settings.
- Designed and oversaw containers with suitable access restrictions.
- Understood SAS tokens that have been generated for safe, transient resource access.
- Lifecycle management was put into practice to automate data movement.
- Understood resource cleanup as a key component of cost control.
- Gained practical knowledge of the key components of Azure storage.

These lessons learned will be useful in the design and management of safe, highly accessible, and reasonably priced cloud storage solutions for next projects or practical uses.

References:

Azure Storage Accounts - How to Create new Storage Accounts. (n.d.). Wwww.youtube.com.
<https://www.youtube.com/watch?v=sI2ahWX8RH8>

CBT Nuggets. (2023, January 19). *How to Generate Shared Access Signatures with Azure Storage Accounts.* YouTube. <https://www.youtube.com/watch?v=0PX1eW1sCGg>

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